

Peer Review: Hierarchical Modelling and Testing

Reviewer: Jenny Laberg Nilsson

Author under review: Kadri Kalamäe

Date: October 2026

- The report starts off with a good exploratory data analysis using a tree diagram to show the variance across datasets and compare sample sizes, which makes for a great start and nicely justifies using a hierarchical model right from the raw data. I specifically like how they add the reuse percentages on the left side of the plot to showcase the variation in the baseline data.
- They continue to explain the model setup very clearly in the text alongside the math formulas under a theory section, which was part of the instructed outline. They use Bambi to display the implementation and default priors as recommended.
- To fit the model and adjust these priors, they plot the prior probability of reuse to motivate their changes. Showing the code while commenting on why they tighten sigma or adjust target accept makes the whole process super transparent. I like how they constantly justify their choices by going back to plot the model parameters. However, for the `az.plot_trace` plots, they could tweak the dimensions so the titles are easier to read and don't overlap the figures. Never the less, the trace plots look well-mixed, indicating good MCMC chain convergence without signs of divergence.
- After deciding on priors, they summarize the parameters in a very readable table as asked for, including 90% credible intervals.
- They go on to present their hypothesis for the effectiveness of the intervention. I do think it is very nice that they convert the raw model output into direct probabilities and percentage effects. I also like that the author points out that moderate heterogeneity cannot be entirely ruled out due to having only seven studies.
- Overall, the report is easy to read with a natural flow. It follows a clear red line that starts with raw data visualization, presenting the model, adjusting and justifying parameters, performing hypothesis testing, and finishing with a short conclusion that wraps up everything they did and the main results. A suggestion that could have been added is a comparison of the raw data to the simulated data, like a PPC.